BLUESTOCK FINTECH

Mutual Fund Analytics Capstone Final Project Report

End-to-End Data Analytics Platform for Indian Mutual Funds

Intern	Akshaya Sada
Track	Data Analysis
Duration	Jun 2026 (7 Days)
Datasets	10 CSVs | 46,000+ NAV Records | 32,778 Transactions
Tools	Python | SQLite | Power BI | Jupyter | Git
GitHub	github.com/akshayasada38/bluestock-mf-capstone

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1. Executive Summary

This report presents a complete end-to-end mutual fund analytics platform developed as part of the Bluestock Fintech Data Analysis Internship Capstone. The project analyses the Indian mutual fund ecosystem across 40 schemes from 10 leading AMCs using 10 structured datasets spanning January 2022 to May 2026.

The platform ingests, cleans, and loads 46,000+ NAV records, 32,778 investor transactions, and 8,050 benchmark index points into a SQLite star-schema database. It computes 7 performance metrics per fund (CAGR, Sharpe, Sortino, Alpha, Beta, VaR, Max Drawdown), builds a composite fund scorecard, and delivers insights through 15 EDA charts and a 4-page interactive Power BI dashboard.

Rs.62.74L Cr	Rs.31,002 Cr	26.12 Cr	32,778
Total AUM	SIP ATH	Total Folios	Transactions

Key Conclusions:

- SIP inflows grew 169% from Rs.11,517 Cr (Jan 2022) to Rs.31,002 Cr (Dec 2025) — a historic all-time high.
- Mirae Asset Large Cap Fund topped the composite scorecard (score 85.9/100, Sharpe 1.45, 3yr CAGR 34%).
- Direct plans consistently outperform Regular plans by 1-2% annually due to lower expense ratios.
- SBI Mutual Fund dominates with Rs.12.5L Cr AUM — nearly 20% market share among 10 AMCs studied.
- 97.8% of investors with 6+ SIP transactions show irregular payment patterns (avg gap > 35 days).
- Large Cap funds are highly correlated (0.85-0.95) — diversification across categories is essential.

2. Data Sources & Description

The project uses 10 structured CSV datasets provided by Bluestock Fintech, supplemented by live NAV data fetched via the mfapi.in REST API. All datasets cover January 2022 to May 2026.

#	File	Rows	Cols	Description
01	fund_master.csv	40	15	Scheme metadata: AMC, category, risk, expense ratio, manager
02	nav_history.csv	46,000	3	Daily NAV per scheme — date, amfi_code, nav
03	aum_by_fund_house.csv	90	5	Quarterly AUM per AMC in Rs. lakh crore
04	monthly_sip_inflows.csv	48	6	Monthly SIP inflows, accounts, YoY growth
05	category_inflows.csv	144	3	Monthly net inflows by fund category
06	industry_folio_count.csv	21	6	Quarterly total industry folios
07	scheme_performance.csv	40	19	Returns, Sharpe, alpha, beta, expense ratio
08	investor_transactions.csv	32,778	13	SIP/Lumpsum/Redemption with demographics
09	portfolio_holdings.csv	322	8	Equity fund sector weights
10	benchmark_indices.csv	8,050	3	NIFTY50, NIFTY100, BSE SmallCap daily data

Live API: Live NAV for 6 schemes fetched via mfapi.in (GET https://api.mfapi.in/mf/{code}). 16,243 records saved as nav_all_schemes_combined.csv.

3. ETL Pipeline Design

3.1 Architecture

The ETL pipeline runs in 6 sequential steps managed by run_pipeline.py as the master execution script.

Step	Script	Action	Output
1	data_ingestion.py	Load 10 CSVs, inspect shape/dtypes, flag anomalies	Console report
2	live_nav_fetch.py	Fetch live NAV from mfapi.in for 6 schemes	6 raw NAV CSVs
3	fund_exploration.py	Explore fund master, validate AMFI codes	data_quality_summary.txt
4	etl_pipeline.py	Clean all CSVs, load into SQLite DB	bluestock_mf.db (23 tables)
5	compute_metrics.py	Compute CAGR/Sharpe/Alpha/VaR/Scorecard	5 metric CSVs
6	recommender.py	Fund recommendation by risk appetite	Console output

3.2 Data Cleaning

- NAV History: Parsed dates, sorted by amfi_code+date, forward-filled weekends/holidays using resample('B').ffill(), removed duplicates, validated NAV > 0, computed daily_return.
- Transactions: Standardised transaction_type, validated amount_inr > 0, fixed date formats, uppercased KYC status.
- Scheme Performance: Validated numeric columns, filtered expense_ratio_pct to 0.1-2.5% range.
- All CSVs: Removed duplicates, stripped whitespace from string columns.

3.3 SQLite Star Schema

Table	Type	Rows	Key	Description
dim_fund	Dimension	40	amfi_code	Scheme metadata
dim_date	Dimension	1,461	date_id	Date spine 2022-2025
fact_nav	Fact	46,000	amfi_code+date	Daily NAV
fact_transactions	Fact	32,778	transaction_id	Investor transactions
fact_performance	Fact	40	amfi_code	Performance metrics
fact_aum	Fact	90	fund_house+month	AUM data

4. EDA Key Findings

15 EDA charts were produced across 10 themes. The most significant findings are:

Finding 1: NAV Trend

Large Cap Direct funds gained 20-35% in 2023 bull run. Market correction Jun-Oct 2024 caused 8-15% drawdowns. HDFC Top 100 and Mirae Asset Large Cap were top performers.

Finding 2: AUM Growth

Total AUM grew 62% from Rs.38.7L Cr to Rs.62.74L Cr. SBI MF leads at Rs.12.5L Cr (20% share).

Finding 3: SIP Inflow

169% growth from Rs.11,517 Cr to Rs.31,002 Cr (2022-2025). Dec 2025 is all-time high.

Finding 4: Category Inflows

Liquid funds attracted most net inflows (Rs.4.51L Cr). Large/Mid Cap show strong Q4 seasonality.

Finding 5: Demographics

26-45 age group: 55%+ of SIPs. Female investors: 38%. 56+ group: highest avg SIP (Rs.14,200).

Finding 6: Geography

MH+KA+DL = 45% of investments. T30 cities: 71% of inflows vs B30: 29%.

Finding 7: Folio Growth

Folios doubled: 13.26 Cr to 26.12 Cr (Jan 2022 to Dec 2025) — +97%.

Finding 8: Correlation

Large Cap pairwise returns: 0.85-0.95 correlation. Category mixing essential for diversification.

Finding 9: Sector Allocation

Banking 28%, IT 18%, Consumer Goods 12% — top 3 sectors in equity fund portfolios.

Finding 10: Expense Ratio

Direct plans avg 0.42% vs Regular 1.18%. Over 10 years, 0.76% difference = 8-12% lower corpus.

5. Performance Analytics

5.1 Metrics

Metric	Formula	Rf	Purpose
CAGR	(NAV_end/NAV_start)^(1/n)-1	N/A	Annualised return
Sharpe	(Rp-Rf)/Std(Rp)*sqrt(252)	6.5%	Risk-adjusted return
Sortino	(Rp-Rf)*sqrt(252)/Downside_Std	6.5%	Downside risk
Alpha	OLS intercept*252 vs NIFTY100	N/A	Excess return
Beta	OLS slope vs NIFTY100	N/A	Market sensitivity
VaR 95%	5th percentile daily returns	N/A	Worst 5% day loss
Max DD	min(NAV/cummax-1)	N/A	Peak-to-trough decline

5.2 Top 10 Scorecard (30% CAGR + 25% Sharpe + 20% Alpha + 15% ER + 10% MaxDD)

Rank	Fund	3yr CAGR	Sharpe	Alpha	VaR	Score
1	Mirae Asset Large Cap	33.99%	1.45	8.2%	-1.02%	85.9
2	ICICI Pru Midcap	31.77%	1.18	6.8%	-1.45%	81.8
3	Kotak Flexicap	29.58%	1.31	5.9%	-1.18%	81.5
4	HDFC Midcap Opp	32.43%	1.09	7.1%	-1.52%	80.3
5	ICICI Pru Bluechip	32.48%	1.03	6.4%	-1.21%	79.5
6	Axis Midcap	35.10%	0.98	9.1%	-1.68%	78.2
7	SBI Bluechip	30.45%	1.02	5.5%	-1.19%	77.8
8	Mirae Tax Saver	29.17%	1.11	5.2%	-1.31%	76.4
9	ABSL Frontline	28.96%	0.97	4.9%	-1.28%	75.1
10	Nippon Large Cap	27.85%	0.94	4.3%	-1.34%	73.6

6. Advanced Analytics

6.1 VaR & CVaR

Historical VaR (95%) was the 5th percentile of daily returns. CVaR is the mean below the VaR threshold. ICICI Pru Liquid: safest (VaR -0.02%). SBI Small Cap: riskiest (VaR -2.69%). Confirms risk category labels align with actual distributions.

6.2 Rolling 90-Day Sharpe

All top 5 funds showed Sharpe > 1 during 2023 bull run, dipped below 0 in Jun-Oct 2024 correction, and recovered by Q1 2025. Computed as returns.rolling(90).mean()/std()*sqrt(252).

6.3 Cohort Analysis

2024 cohort is largest (4,624 investors, avg SIP Rs.10,997). 2022 cohort shows higher avg SIP (Rs.13,400+) as experienced investors increase commitments over time.

6.4 SIP Continuity

97.8% of 3,271 investors with 6+ SIPs have avg gap > 35 days (at-risk). Median gap: 38 days vs expected 30 days for monthly SIPs.

6.5 Sector HHI

Axis Bluechip: highest HHI (0.2064 — concentrated). SBI Small Cap: most diversified (0.1073). Higher HHI funds show higher tracking error vs benchmarks.

7. Dashboard Overview

The Power BI dashboard has 4 pages with slicers for dynamic filtering:

Page 1 — Industry Overview

KPI cards + AUM trend line + AUM by AMC bar. Slicers: Year, Fund House.

Page 2 — Fund Performance

Return vs Risk scatter + Fund scorecard table + NAV vs Benchmark. Slicers: Fund House, Category, Plan.

Page 3 — Investor Analytics

Transaction by state + SIP/Lumpsum/Redemption donut + Age group bar + Monthly volume. Slicers: State, Age Group, City Tier.

Page 4 — SIP & Market Trends

Dual-axis SIP + NIFTY50 + Category inflow matrix. Slicers: Year, Category.

8. Limitations

- Scope limited to 40 schemes — not representative of all 1,908+ AMFI-registered schemes.
- Transaction data is sample/synthetic — real behaviour may differ significantly.
- CAGR uses actual trading days vs SEBI-mandated business-day calculations.
- Power BI requires manual refresh — no real-time API connection.
- Recommender is rule-based — ignores investor horizon and tax implications.
- Sector HHI uses single-point-in-time holdings; actual holdings change quarterly.

9. Recommendations

For Investors

- Choose Direct plans — save 0.76-1.2% annually in expense ratio.
- Diversify across Large Cap, Mid Cap, and Debt categories.
- Automate SIP payments to avoid the 38-day average irregularity gap.
- Consider Mirae Asset Large Cap or Kotak Flexicap for best risk-return balance (Sharpe > 1.3).

For AMCs

- Focus on B30 city penetration — only 29% of inflows from Tier 2/3 cities.
- Implement SIP reminder systems to address 97.8% at-risk rate.
- Increase transparency on quarterly portfolio holdings disclosure.

For the Platform

- Schedule mfapi.in refresh via cron job (8 PM weekdays).
- Extend to all 1,908 AMFI schemes.
- Add Monte Carlo simulation for NAV projections.
- Integrate LTCG/STCG tax-adjusted returns.

10. Self-Review Checklist

#	Objective	Status	Evidence
1	ETL pipeline runs without errors	PASS	run_pipeline.py — all 6 scripts complete
2	SQLite DB with correct schema	PASS	bluestock_mf.db — 23 tables, star schema
3	EDA notebook 15+ charts	PASS	EDA_Analysis.ipynb — 15 charts
4	Performance metrics — correct formulas	PASS	Performance_Analytics.ipynb
5	Dashboard — 4 pages + slicers	PASS	bluestock_mf.pbix
6	Advanced analytics notebook	PASS	Advanced_Analytics.ipynb
7	Final report (15-20 pages)	PASS	Final_Report.pdf — 16 pages
8	Presentation (12 slides)	PASS	Bluestock_MF_Presentation.pptx — 12 slides
9	GitHub + README + v1.0 tag	PASS	github.com/akshayasada38/bluestock-mf-capstone
10	Code runs without errors	PASS	Tested on Python 3.13 + Windows 11

Project Status: COMPLETE — All 10 objectives met

Bluestock Fintech Capstone | Akshaya Sada | Jun 2026